

1)

⇒ Descriptive → Works on the history of data.

Use case: Suppose 3000 cups of tea was sold in the weekdays and on the weekend 4000 cups of tea was sold. This draws conclusion on why the has much revenue generated and so on.

Prescriptive → What should we do.

Use case: In Airline the if the ticket is overbooked. Then based on passenger's condition like: history, class, weighs it reroutes the passengers.

Predictive → What will happen.

Use case: A credit card company make the card expire in 90 days. The company makes the prediction of customer maximum usage based on his/her past expenditures data.

Diagnostic analysis $\rightarrow$ Why did it happen.

use case: If your shop goes on loss back to back and. Then why is it happening. May be because of the new shop opening, maybe out of trends, etc.

2)

$\Rightarrow$

Exploration: It is the act of trying out novel actions that the agent has little to no experience with. The reinforcement learning tries to maximize the reward observing the environment. It's goal is to discover new states, understand the environment and find potentially higher rewards to the that are currently unknown.

Exploitation: It means using the information explored. It chooses the best of the according to data explored, and if environment contains any hidden treasures or better paths that agent hasn't tried yet, it will never find them.

3)

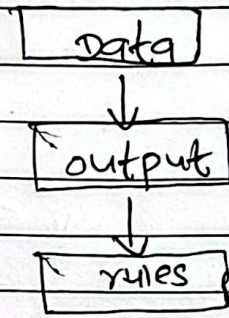


fig : Basic workflow process.

The ML basic workflow is it collects data. after the problem formulation is done. Then based on that it gives outputs and it later find the rules / algorithms it is working on.

For eg: You feed the computer of thousands of past transactions which is (data) and label which ones were actually scammy (outputs). The machine Learning algorithm looks at both and generates its own complex formula (Rules).